

James Gabriel Elijah Ty

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EDUCATION

University of The Philippines Cebu

BS Computer Science

Expected: July 2027

Gorordo Avenue, Labug, Cebu City, 6000 Philippines

UTokyo Virtual Exchange Program - CALL (Computer Assisted Language Learning)

Apr. 2026 – Jul. 2026

- Evaluated a VR language learning application (NounTown) using SLA theories and proposed a custom CALL framework integrating Generative AI, VR, and secondary language acquisition concepts.

PROFESSIONAL EXPERIENCE

ISCOLab - Research / Developer Intern

Jan. 2026 – Jul. 2026

- Tech: C#, Unity, Blender, OpenXR, WebXR, A-Frame, Frame VR**
- Implemented Level of Detail optimizations for the UPOU metaverse, improving performance by 63%.
- Created a webXR based semiconductor inspection training prototype, which was chosen to be further developed as a microcredential for a Taiwanese semiconductor company.
- Presented demonstrations on the application of XR technologies for Japanese language learning and instructor training to the Japan Foundation, which was well received.

PROJECTS

CreditPass: An AI-Driven Capacity Profiling & Containerized Credit Scoring System

- Tech: FastAPI, Next.js, TensorFlow, Docker, Huawei Cloud, Redis**
- Engineered an end-to-end, privacy-preserving credit capacity profiling engine utilizing alternative data structures like telco and e-wallet telemetry.
- Architected a resilient cloud infrastructure on Huawei Cloud, routing client traffic through ELB and Cloud CDN to containerized microservices hosted on Cloud Container Engine (CCE).
- Designed an automated MLOps retraining pipeline that monitors model drift via Cloud Eye and orchestrates continuous model retraining and evaluation workflows through ModelArts.

Sabot: A Blockchain Escrow & Decentralized Transaction Verification Platform

- Tech: Next.js, TypeScript, Solidity, Hardhat, Lisk Sepolia, Supabase, Tailwind, OpenZeppelin**
- Developed a composable peer-to-peer transaction safety layer featuring immutable agreement lifecycles and automated multi-deliverable escrow functionality.
- Co-authored and deployed Solidity smart contracts managing token economics (SBT), automated platform/arbitrator fee splits, and decentralized dispute resolution workflows.
- Implemented an oracle-based cryptographic verification system to automatically validate file-based and service-oriented deliverables securely on-chain.

LakbAI: A Dynamic Pedestrian Accessibility & Geospatial Decision-Support Platform

- Tech: Next.js, TypeScript, Django, Zustand, Leaflet, AWS, Turf.js, PyTorch**
- Developed a geospatial dashboard to visualize and assess urban pedestrian accessibility across pilot areas in the Philippines using a dynamic fuzzy logic inference system.
- Integrated multi-modal data streams including real-time weather APIs, flood hazard models, OpenStreetMap infrastructure data, and manual street audits into a continuous [0, 1] scoring model.
- Built an interactive GeoJSON mapping interface with 13-band color interpolation to deliver at-a-glance, actionable localized insights for city planners and policymakers.

GreenPoint: A GIS-Based Urban Greening Framework

- Tech: Next.js, Supabase, Prisma, Mapbox, Leaflet, Google Earth Engine, LangGraph, AWS**
- Engineered a Geographic Information System (GIS) and urban planning platform to identify, evaluate, and recommend site-specific greening interventions in Mandaue City.
- Integrated high-resolution satellite imagery (NDVI, LST) and air quality data to compute a multi-dimensional Greenery Index focused on quantity, equity, and connectivity.
- Developed an AI recommendation engine using LangGraph agentic workflows and an urban forestry RAG system to generate strategies like pocket parks and blue-green corridors.

Wafers, Please!: A WebVR Semiconductor Inspection Prototype

- Tech: Next.js, TypeScript, A-Frame, WebXR**
- Developed an XR-style educational training prototype simulating a beginner semiconductor inspection shift loop.
- Implemented deterministic defect generation alongside dual progression gates evaluating operational performance and decision quality.

ACHIEVEMENTS AND AWARDS

Huawei Developer Competition 2025 (CreditPass)

2nd Runner Up

APRU Tech Policy Hackathon 2025 (CreditPass)

Honorable Mention

Lisk Builders Challenge Round 3 (Sabot)	Honorable Mention
Philippines Junior Data Science Challenge 2024 (LakbAI)	Champion
University of the Philippines Cebu - EXPOdition 30 (GreenPoint)	Project of the Year
27th Philippine Statistics Quiz 2023	Regional Champion
CESAFI Computer Quiz Bowl	2nd Runner Up (College 2026) Champion (HS 2023)
UPCSG Inter-High School Programming Competition	1st Runner Up

CERTIFICATIONS

AWS Certified - Solutions Architect Associate	Aug. 2026
AWS Certified - Cloud Practitioner	Aug. 2026
AWS Certified - AI Practitioner	Aug. 2026
Coursera - AWS Cloud Support Associate Professional Certificate	June 2026
Coursera - AWS Cloud Technology Consultant Professional Certificate	May 2026
AWS AI Practitioner Challenge	Nov. 2025
Huawei Cloud Developer Certification - HCCDA AI	Nov. 2025
Huawei Cloud Developer Certification - HCCDA Tech Essentials	Nov. 2025

SKILLS AND TECHNOLOGIES

Languages and Frameworks:	C#, TypeScript, Go, Python
Frontend and Backend:	Next.js, Express.js, FastAPI, Django, Fiber, React Native, A-Frame
Databases and ETL Tools:	PostgreSQL, MySQL, Supabase, MongoDB, Redis
Cloud and CI/CD tools:	AWS, GCP, Huawei Cloud, Jenkins, GitHub Actions, Terraform, Docker
Machine Learning and AI:	TensorFlow, AWS SageMaker, n8n, Computer Vision, RAG

LEADERSHIP EXPERIENCE

GitHub Campus Experts - Campus Expert - Skills: Community Analysis, Technical Advocacy, Inclusion Strategy, Strategic Communication - Selected as a campus leader through GitHub’s rigorous selection process with a 1% acceptance rate. - Formulated a comprehensive Community Assessment and Inclusion Framework for the UP Cebu tech ecosystem. Designed strategic methodologies to lower technical barriers to entry and mitigate the regional digital divide. - Championed inclusive community spaces by enforcing a structured Code of Conduct, designing asynchronous learning pathways, and defining accessibility guidelines for regional developer events.	Oct 2025 – Present
AWS Student Builder Group UP Cebu - Lead - Skills: Technical Leadership, Cross-Functional Coordination, Project Management, Cloud Advocacy - Pioneered the campus cloud community launch and established learning pathways using AWS Skill Builder, guiding peers through hands-on projects with AWS Lambda, API Gateway, S3, and Amazon RDS. - Spearheaded the collaborative organization of AWS Community Day Cebu 2025, coordinating a cross-functional coalition of 8 regional universities to deliver a successful conference for 200+ cloud enthusiasts.	Nov 2025 – Present
UPCSG - Executive Director - Organizational Leadership, Strategic Planning, Financial Resourcefulness, Stakeholder Management - Elected to govern UP Cebu’s premier computer science organization, overseeing the end-to-end execution of flagship academic programming, hackathons, and developer workshops. - Directed the UPCS Hackathon 2025 and 2026, driving inter-collegiate collaboration to bring together 50 total team applications with 15 finalists across 10 different schools. - Managed, mentored, and empowered a cross-functional workforce of 75+ student volunteers.	June 2025 – June 2026
UPCSG - Education and Development Director - Skills: Curriculum Design, Technical Training, Event Organization, Public Speaking - Designed and executed targeted educational roadmaps, managing Upskilling Workshops that regularly equipped 50+ students with industry-relevant software engineering concepts. - Restructured and managed a comprehensive Peer Tutoring Program serving ~50 students; formulated custom lesson plans, created rigorous mock evaluations, and personally conducted academic tutoring sessions. - Served as Co-Head for the Scratch That! outreach initiative, designing accessible programming curriculums to introduce foundational computer science concepts to 60 local high school students through Scratch.	June 2024 - June 2025

REFERENCES

Roberto Figueroa Jr. <i>XR Researcher, Lab Head, ISCOLab</i> rfigueroa@upou.edu.ph	Cyrus Pastelero <i>Senior Software Developer, Reap</i> cyrus@pastelero.ph
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